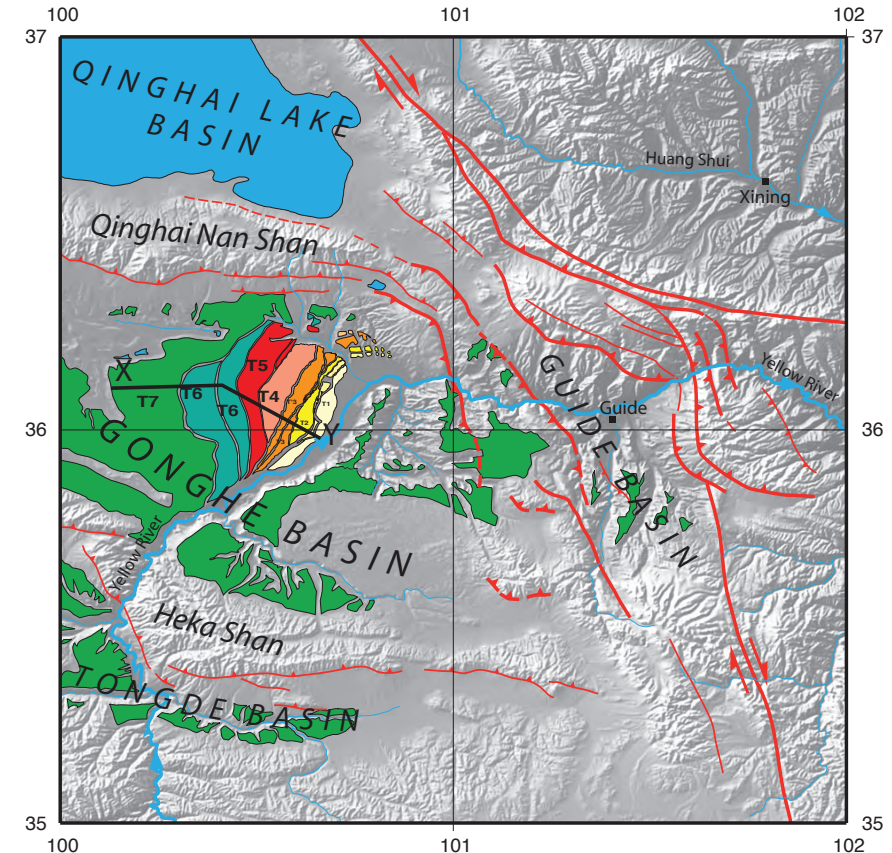
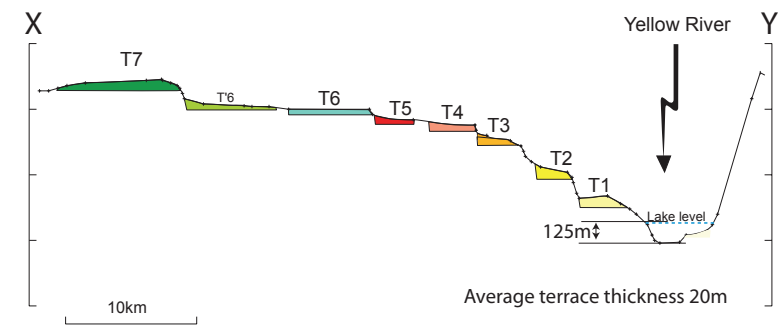


The google earth image below shows a section of the Yellow River, near Qinghai Lake in NE Tibet. The region is located between two major left-lateral strike-slip faults, the Kunlun Fault to the south and Haiyuan Fault to the north. Tectonic activity likely began at the end of the Miocene or at the beginning of the Pliocene.



The yellow pins on the google earth image show locations of 7 river terraces that record the incision history of the Yellow River in this area. The cause of incision is not clear, it may represent the upstream extent of a transient wave of incision that initiated near the topographic margin of the Tibetan plateau, or it may relate to local deformation driven by active faulting and/or resistant lithology. We cannot begin to explain the cause until we have a basic chronology that defines the age of incision and its rate. **The aim of this exercise is to define the incision history for this area.**

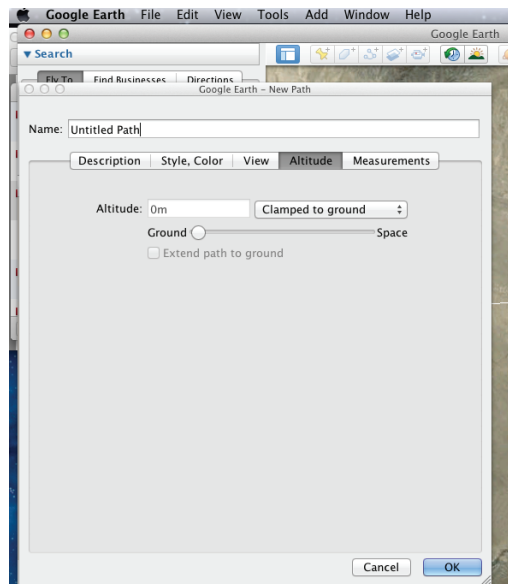


You will use google earth to measure the amount of incision for each terrace.

The first step is to locate the study area. Download the google earth .kmz file *called “elevation profile”* open google earth and the kmz file. This will take you to the study area.

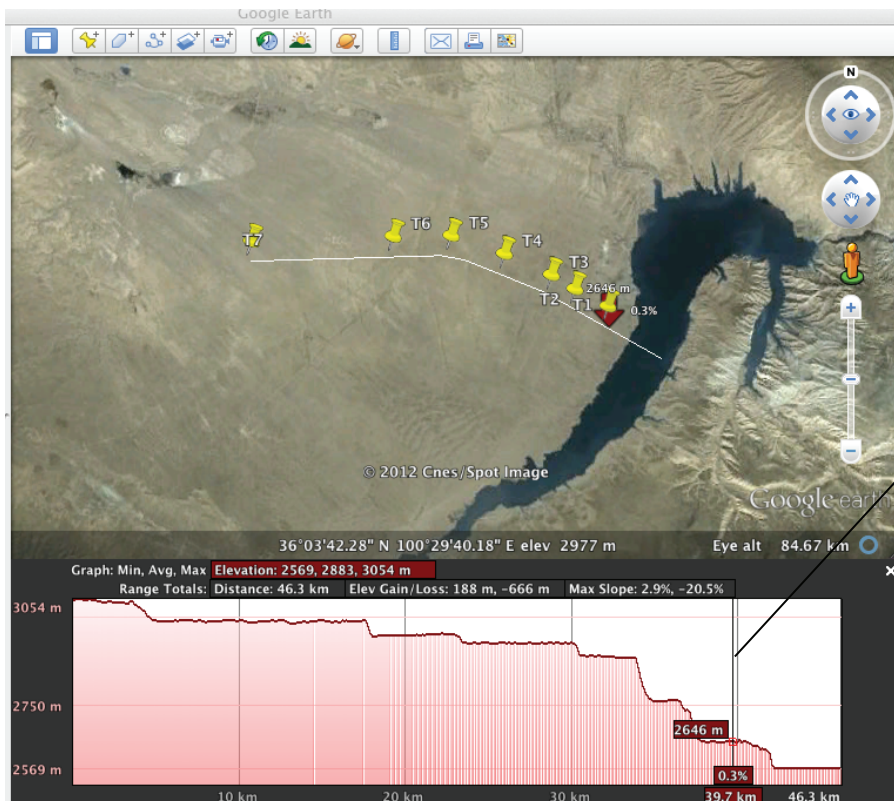
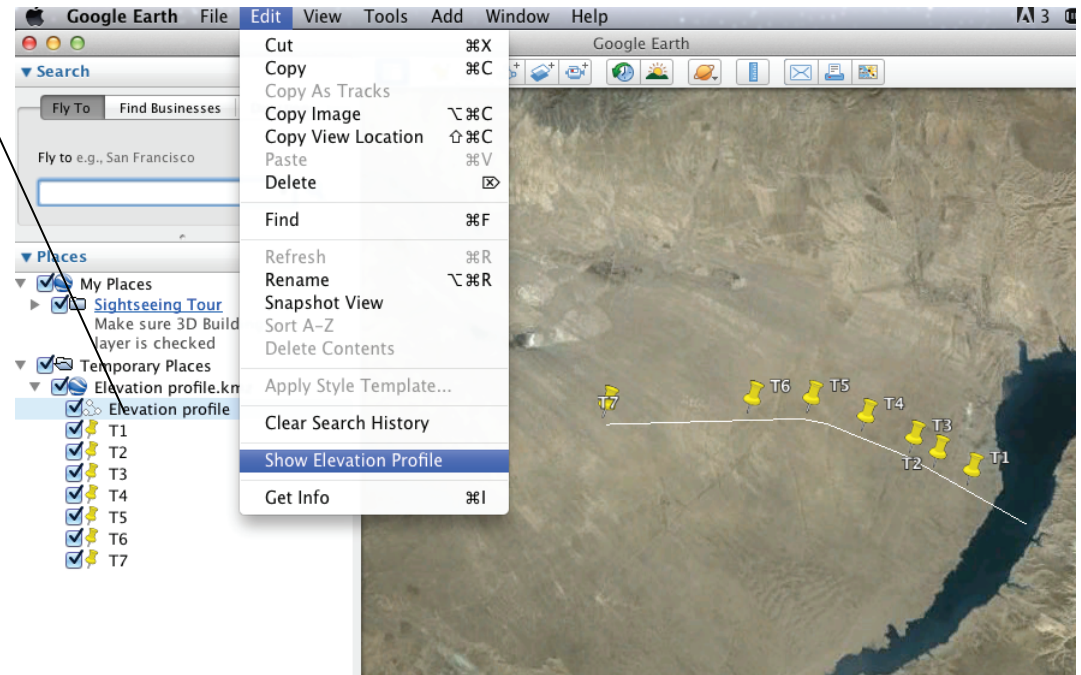
You will see the yellow pins for the 7 terraces and a white line, which is the path along which an elevation profile can be plotted.

If you want to draw an elevation profile across any area you will need to select **altitude** in the path settings box as shown here.



To see the elevation profile for any path select the object “Elevation profile”

Next in the menu select Edit and choose the option “show elevation profile” as shown on the right



Use the mouse to drag a line on the graph and this will give you readings of location and elevation for any given point along a selected path. We will use this feature to measure the elevation change (amount of incision) between each ancient river terrace.

**Task 1:** Record the elevation for each terrace and plot the data. Start from the lake which has a depth of 22m (the bed is the current river incision base point).

**Question 1:** Has incision been constant?

The paleo river beds are covered by stranded river sediment deposited during an aggradational phase and are known as strath terraces. Sediment is deposited when river water supply was low or when sediment supply exceeded river transport capacity. These deposits are preserved due to the river returning to an erosional phase cutting down into bedrock (due to reduction in sediment supply or increase in water supply and/or base level change). To define an incision rate we need to know;

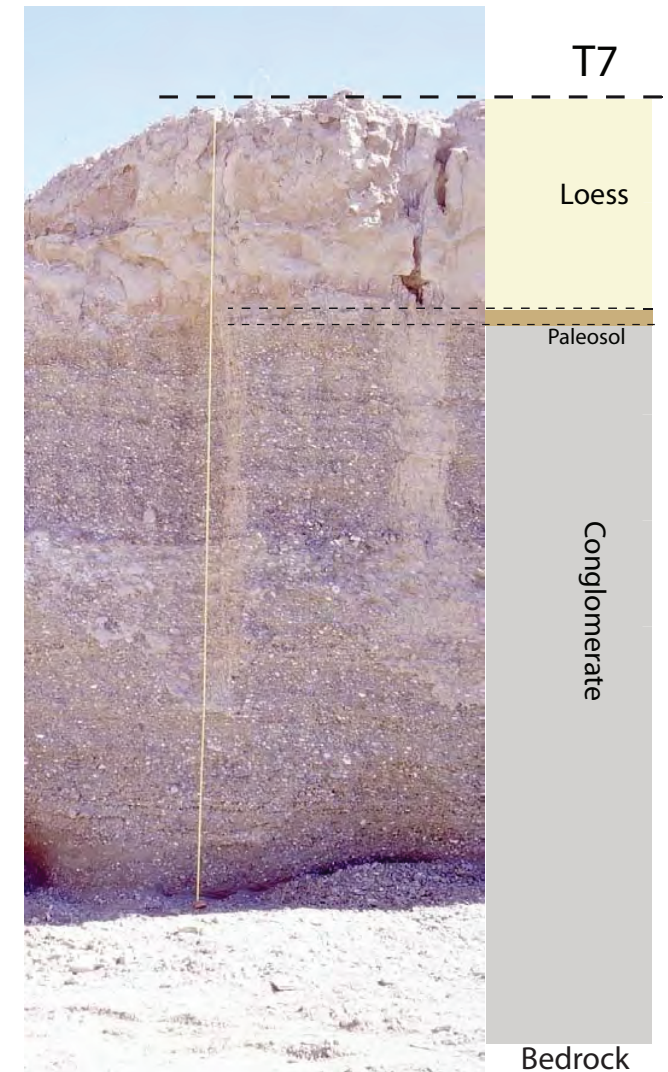
- 1) Amount of incision between each terrace
- 2) Deposition age for each terrace

The amount of river incision is the height difference between the bedrock river bed on which the terrace sediments were deposited so we need to subtract the thickness of each terrace deposit (and any soil or later deposit sitting on top) from the surface elevation that we measure on google earth.

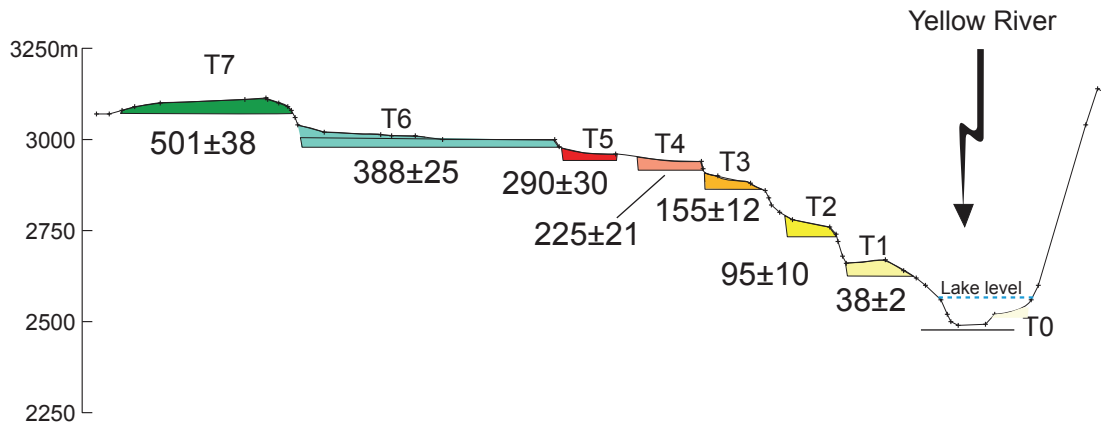
The photo on the right shows a section of terrace 7. The terrace deposit is capped by a paleosol, which marks the end of river aggradation and loess deposits sit on top. If we assume that the terrace deposits were deposited over a short period of time of the paleosol. Age dating the paleosol would give a maximum age (cannot be older than) for when the river ended an aggradational phase and therefore when the river entered an erosive phase cutting down into the bedrock (incision).

Dating of Quarternary or Holocene paleosols can be achieved by using either radio-carbon methods on organic material (wood or shells) or by cosmogenic radionuclide methods known as exposure dating. We will learn about these methods in a later lecture. For the purposes of this exercise the methods used to obtain the mean ages of the terrace deposits (see next page) is not of importance.

Clearly google earth cannot be used to measure the thickness of the terrace deposits and any cover sediments/soil (only field mapping can do this) so for the purposes of this exercise we will assume each terrace deposit has an average thickness of 20m.



The ages for each terrace deposit is provided below in units of thousand years (Ka).



**Task 2:** Calculate the average incision rate (in units of mm/yr) for any group(s) of terraces that show similar rates of incision.

**Question 2:** There is no change in rock type over the area so we can rule out changes in erodability as a cause for terracing. This leaves two possibilities, either tectonics (base level changes) or climate (change in precipitation rate and therefore change in river power and sediment transport capacity). The plot on the right shows the local east asian monsoon climate record. Use this information together with your plots of incision rate to decide if climate changes had a role in terracing.

The plot below shows  $\delta^{18}\text{O}$  which is a measure of the ratio of stable isotopes  $^{18}\text{O}:^{16}\text{O}$ . It is a good proxy for climate in that it is a measure of the temperature of precipitation because  $^{16}\text{O}$  becomes enriched with preferential evaporation from seawater. For climate records researchers use carbonate cave deposits as they are assumed to record the intensity of monsoon precipitation because the  $\delta^{18}\text{O}$  of the carbonate tracks the isotopic signature of precipitation.

The plot displays a broadly sinusoidal pattern that reflects a strong influence from orbital forcing, with a well developed 23-26 kyr cycle related to precession. Superimposed on this are trends related to glaciations. With increased monsoon intensity there is an increase in rainfall so rivers will have greater erosiveness or transport capacity.

